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Samuel T. Woodrich for the degree of Master of Science in Environmental Sciences presented on March 16, 2022.

Title: An Observational Study of the Effects of Wildfire on Hermit and Varied Thrush Distribution in the Mt. Washington Wilderness Area

Abstract approved: _____

Joan C. Hagar

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An Observational Study of the Effects of Wildfire on Hermit and Varied Thrush Distribution in the Mt. Washington Wilderness Area

by

Samuel T. Woodrich

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APPROVED:

Major Professor, representing the Department of Forest Ecosystems & Society

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Director of the Environmental Sciences Graduate Program

Dean of the Graduate School

I understand that my thesis will become part of the permanent collection of Oregon State University libraries. My signature below authorizes release of my thesis to any reader upon request.

Samuel T. Woodrich, Author

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CONTRIBUTION OF AUTHORS

My colleagues John (Yanni) Bouranis and Adrienne Lurada contributed their time and expertise throughout the development of this study by assisting me write code necessary to evaluate my data and by providing me with feedback and edits to my paper respectively. Dr. Joan Hagar contributed to the development of this paper by mentoring me as well as providing feedback and edits to each draft written. Dr. Steve Acker provided code used in his original vegetation analysis as reference for developing new code for this study as well as feedback on my drafts. Finally Dr. Mark Schulze provided additional feedback and edits on my drafts.

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DEDICATION

I dedicate this thesis to my father, without whom I wouldn't be the person I am today. The mutual interest that you have developed for how the natural world works, the writing that you have indulged yourself in organically, and your continued commitment to sharing your discoveries with me have continuously propelled my drive to learn more and share every experience with you. Words cannot describe the amount of happiness that sharing this achievement with you will bring me. I love you very much.

An Observational Study of the Effects of Wildfire on Hermit and Varied Thrush Distribution in the Mt. Washington Wilderness Area

Abstract

The hermit thrush (*Catharus guttatus*) and the varied thrush (*Ixoreus naevius*) are focal species for conservation of forested habitats in Oregon. Although habitat associations have been defined for both species, their response to changes in vegetation following wildfire has not been well studied. To help fill this information gap, I used data from remote audio recordings to compare detection frequencies of both species between burned and unburned areas in the Mount Washington Wilderness, Oregon, USA. I also compared key habitat variables for each species between burned and unburned areas. My results show that hermit thrush occurrence was greater in unburned plots and may be influenced by live tree basal area and snag density. I did not find evidence of a difference in varied thrush occurrence between burned and unburned plots. However, varied thrush occurrence may be correlated with canopy cover and live tree basal area following a fire. Further investigation of the effects of mixed severity fire is necessary in future studies to better understand the influence fire has on both thrush species occurrence in the high Cascades.

Chapter One: Introduction

A majority of avian species in North America have experienced a decrease in population approaching a net loss of 3 billion birds within the past 48 years as a result of habitat loss, climate change, unregulated harvest, and other forms of human-caused mortality (Rosenberg et al. 2019). The effects of more frequent and severe fire disturbances related to climate change (Westerling et al. 2006, Yue et al. 2013, Wing and Long 2015, Reilly et al. 2017) on native passerine species in the high Cascades are not fully understood. Improved understanding of the effects that fire disturbance has on forest species and their habitat could inform efforts to preserve natural habitat and prevent further population declines of species.

Partners in Flight (PIF), an active avian conservation group in North America provides bird conservation plans to land managers outlining specific habitat requirements and characteristics that are associated with avian species in order to provide managers with a tool for supporting habitat management and conservation planning. Oregon Partners in Flight outlined a set of avian focal species that represent avian communities throughout Oregon, including habitats west of the Cascades (Altman and Alexander 2012). The conservation plan identifies avian focal species that require key habitat components that represent a functioning ecosystem and thus, represent many other bird species and other elements of biodiversity (Chase and Geupel 2005, Stephens et al. 2019).

In the Cascades, the varied thrush (*Ixoreus naevius*) has been identified as a focal species for late successional coniferous forests by Oregon Partners in Flight (Altman and Alexander 2012). While the species is listed as “least concern” (Global IUCN Red List) and has an estimated global population size of 35 million individuals (Partners in Flight 2021), local populations in the Cascades are in significant decline (Marshall et al. 2003). Additionally, the varied thrush has been identified as a bird of conservation concern, and is at risk of being listed as an endangered species in the continental United States (USFW 2021). The hermit thrush (*Catharus guttatus*) has been identified as a focal species for mixed conifer/mixed conifer-hardwood forests in the Klamath Mountains where mixed severity fires are frequent (Altman and Alexander 2012). Local populations of the species in the Cascades appear to be stable (Marshall et al. 2003), with the species listed as “least concern” (Global IUCN Red List) and an estimated global population size of 72 million individuals (Partners in Flight 2021). Additionally, the species has increased in population by 14.8% from 2010 to 2019 in North America (Rosenberg et al. 2019). Both the hermit and varied thrush serve as the focal species for this study in the high Cascades.

The hermit thrush is a migratory species that breeds in the high Cascades from mid to high elevations (Marshall et al. 2003) in various coniferous, deciduous and mixed forest types in the United States (Rosenberg 2003). The hermit thrush primarily forages on the ground for insects, but occasionally preys on amphibians and reptiles (Jones and Donovan 1996). There is little information regarding foraging characteristics in Oregon, however there is documentation of hermit thrush feeding on the fruit of blue elderberry (*Sambucus cerulean*), huckleberry (*Vaccinium spp.*), blackberries and thimbleberry (*Rubus spp.*), wild raspberry (*Rubus occidentalis*), and golden currant (*Ribes aureum*) as well as moths (Lepidoptera), stoneflies (Plecoptera), and inchworms (Lepidoptera*: Geometridae*) in Oregon (Marshall et al. 2003). Because it nests on the ground in dense brush or in small trees (Mannan 1980) the hermit thrush is especially vulnerable to fire during the breeding season (Marshall et al. 2003). On the other hand, the promotion of understory growth through the reduction of forest canopy cover following a fire could benefit the hermit thrush (Altman and Alexander 2012). Hermit thrush abundance has been positively correlated with snag density in previous studies (Schwab et al. 2001, Simon et al. 2002, White et al. 2016). However, salvage logging following a fire may negatively influence hermit thrush abundance through the removal of dead and dying trees (Simon et al. 2002), though specific associations between the species and burned snags have not yet been identified. Additionally, studies that have identified a positive correlation between hermit thrush abundance and snag density have been conducted outside of Oregon (Schwab et al. 2001, White et al. 2016). This previously documented correlation may be because snags provide food and a post to sing on following a fire (Simon et al. 2002). Additionally, hermit thrush became less abundant in burned plots as time progressed immediately after a fire disturbance, possibly due to the decreased snag density over time since fire (Schwab et al. 2001).

Studies specifically addressing effects of fire on habitat for songbirds, including hermit thrush, have been conducted across a variety of forest types, sometimes leading to conflicting conclusions about the species response to fire (Kotliar et al. 2007, Kennedy and Fontaine 2009, Gaines et al. 2010). A study involving the response of hermit thrush over time since fire has not been conducted in the high Cascades. However, the promotion of long-term shrub and herbaceous communities in forested areas has also been associated with the species in the Klamath Mountains, where mixed severity fires are frequent (Altman and Alexander 2012). Studies in a variety of forest systems in North America have found that habitat variables such as snag density (Schwab et al. 2001, Simon et al. 2002, White et al. 2016) and shrub cover (Altman and Alexander 2012) are positively correlated with hermit thrush occurrence.

Habitat types used by the varied thrush include structurally complex late successional old growth/mature forest in the high Cascades. The species breeds primarily in forests >60 years old that contain multiple tree layers with mixed deciduous and coniferous composition, berry-producing trees and shrubs, high canopy closure (>70%), and >30% deciduous cover in the sub-canopy and understory (Altman and Alexander 2012). The species preys on invertebrates on the forest floor early during the breeding season and includes more fruit in the diet as berry crops ripen (Beck and George 2000). When ripe, varied thrush diet of fruit-bearing shrubs includes huckleberry (*Vaccinium spp.*), thimbleberry (*Rubus parviflorus*), salmonberry (*Rubus spectabilis*), and manzanita (*Arctostaphylos spp.*) (George 2000, Marshall et al. 2003). The varied thrush is listed as a forest-interior or area-sensitive species due to their requirement for a contiguous old-growth or mature forest habitat that spans a greater area than an individual's home range (16 ha) (Brand and George 2001). Significant declines in the Cascades population have been attributed to habitat fragmentation and loss of mature forest habitat (Altman and Alexander 2012).

The effects of fire on varied thrush habitat within the high Cascades have not been researched thoroughly. This includes but is not limited to the impact that fire disturbances have on populations in the high Cascades and quantification of the amount of time that passes before recolonization of burned sites. Although the effects of forest thinning and habitat loss on varied thrush populations in the Cascades have been fairly well studied (Altman and Alexander 2012), a gap in knowledge in the species response to fire still exists. Prior to this study, habitat features associated with the varied thrush have not been studied in relation to fire disturbance in the high Cascades.

Studies that involve the collection of avian data prove to be costly and time consuming when they are conducted in distant and remote regions. While point count surveys are often valued as a thorough and efficient method for sampling avian data, they can prove to be disadvantageous compared to the deployment of recording devices in some cases. Environmental challenges such as snowpack that persists into the breeding season at high elevations, or wildfire disturbances serve as a barrier for data collection in avian studies that take place in wilderness areas. Additionally, financial restraints may justify the use of bioacoustics for data collection as a viable substitute for point count surveys.

In addition to being accepted as a viable alternative to point count surveys, the use of bioacoustics also has many advantages. The use of recording devices allows researchers to gather data between multiple locations simultaneously without having to hire multiple surveyors. Deploying recording devices also provides researchers the opportunity to collect data within remote areas when

point count surveys may not be feasible. Due to the lightweight nature of some recording devices and the ability for most devices to record for several months without requiring human interaction, recording devices can be placed deep within study areas that may otherwise be inaccessible during periods of the study. Finally, collecting a large number of recordings during a study season can mitigate the effects of environmental factors that influence the detectability of avian species on particular days such as wind and rain (Thomas et al. 2020).

Some limitations that are associated with recording devices as opposed to point count surveys in an avian study involve the inability to consistently identify multiple individuals of the same species in a sample site, and the increased time and effort involved in data processing procedures. The inability to identify multiple individuals of the same species limits count data to presence or absence, as opposed to species abundance. Additionally, the distance that recording devices can detect an avian species varies between devices, sometimes leading to unpredictable variation in detection probability (Darras et al. 2018). This issue can be mitigated through the use of multiple recording devices within the same plot although this technique generates a greater financial cost on the researcher (Celis-Murillo et al. 2009, Hutto and Stutzman 2009). Data processing time is sometimes increased in comparison to point count surveys due to the associated time it takes to playback recordings as opposed to collecting data on the study site (Hutto and Stutzman 2009). Technical difficulties and environmental hazards can also lead to the loss of collected data, either through recording devices malfunctioning or from fauna or other natural disturbances destroying the devices.

Remote recordings were used to collect survey data on our focal species for a variety of reasons. For ecological studies similar to this one, the use of recording devices as a method of sampling songbird species has: provided the researcher with the ability to limit inter-observer error that can occur between point count observers by providing the opportunity for multiple users to identify species detected in the same recording (Celis-Murillo et al. 2009); become an equally cost-effective alternative to the use of point count surveys in some avian studies (Acevedo and Villanueva-Rivera 2006); and reduced human interference with the species being studied (Venier et al. 2012). Multiple studies comparing bioacoustics data collection, including the use of recording devices in comparison to traditional point count data collection methods have established the usefulness of bioacoustics data for avian ecological studies when other survey methods are not practical or appropriate (Celis-Murillo et al. 2009, Hutto and Stutzman 2009, Venier et al. 2012, Darras et al. 2018, Thomas et al. 2020).

In an effort to mitigate the loss of avifauna in North America, a greater understanding of focal species response to fire disturbance and further studies of the effects that fire disturbance has on crucial habitat features associated with avian species is necessary. I compared detection rates of two avian focal species in the high Cascades, the hermit thrush and the varied thrush, between burned and unburned plots in the Mt. Washington Wilderness of Oregon. I also compared key habitat characteristics for each species, as identified in the Oregon Partners in Flight Conservation Plan for western forest birds (Altman and Alexander 2012), between burned and unburned plots to better understand how fire might change habitat for these species in the high Cascades. I selected the varied thrush and hermit thrush as focal species for my study because both species are abundant in the high Cascades during the breeding season, but vary greatly in total number of individuals, population trends, and habitat associations from one another.

In this study, my goal was to identify whether or not hermit and varied thrush occurrence was associated with fire disturbance in the western Cascade region. To satisfy this goal, I addressed two research questions:

1.) Does the occurrence of hermit and varied thrush differ between burned and unburned plots in the Mt. Washington Wilderness Area?

2.) How do key habitat features for hermit and varied thrush differ between burned and unburned forest ~6 years after fire?

Chapter Two: Study Design

Study Area

The study was conducted in the southwest portion of the Mt. Washington Wilderness area in western Oregon. The Mt. Washington Wilderness area encompasses 54,278 acres located in the western region of the high Cascades and is managed by the U.S Forest Service Willamette and Deschutes National Forest. This region within the wilderness area was selected due to the high probability that certain sections within this region would have prescribed fires ignited as part of the USFS's plan to reduce fuel load on the forest floor while other adjacent sections would not (Acker et al. unpublished). Vegetation in the region of the Mt. Washington wilderness area where this study took place predominantly consists of mountain hemlock (*Tsuga mertensiana*) and silver fir (*Abies amabilis*) of

varying successional ages from new growth to old growth (Oregon Biodiversity Information Center 2018, Acker et al. unpublished). Elevation ranged from 4600 to 5500 feet within the study area. The Scott Mountain burn ignited in 2010 and burned 3,454 acres within the Mt. Washington wilderness area (Figure 1).

Study Design

The original study was initiated as an experimental test of the effects of fire using planned burning in a wilderness area. However, due to initial public opposition, and a subsequent lack of ideal conditions for ignition, the fire was never ignited as planned, so it has not yet been possible to collect post-fire data. However, a natural wildfire had burned a portion of the study area in 2010, four years prior to the first data collection for the study, so I used this burned area as a basis of comparison with the unburned area that occupied the majority of the study area (Acker et al. unpublished).

Vegetation data were collected in the original study (Acker et al. unpublished). While the original experiment intended for sampling of vegetation and birds to occur before and after the planned burns took place, the Scott Mountain Fire provided a basis for comparison between burned and unburned conditions. I organized the data into two groups to address my research questions: plots that were located within the boundary of the Scott Mountain fire of 2010 (burned), and plots that were outside the boundary (unburned).

I compared five vegetation characteristics considered to be important to the focal thrush species between burned and unburned plots in this study (Table 1). I added live tree basal area as a predictor variable because it is ecologically related to vegetation characteristics identified as important and is a metric commonly used in forest management.

Table 1.) P.I.F Habitat Characteristics for Hermit and Varied Thrush

Vegetation characteristics associated with hermit and varied thrushes from published sources and source from which the characteristic was derived.

	Associated Focal Species Correlation or Requirements:		Association Identified in:		Ecologically Related to:
Vegetation Characteristic	**Hermit Thrush**	**Varied Thrush**	**PIF (Altman 2012)**	**Other Source**	
Fruit-Bearing Shrub Cover	-	**Positive correlation**	**YES**	-	-
Total Shrub Cover	**30% - 70%**	**>30%**	**YES**	-	Fruit-Bearing Shrub Cover (%)
Live Tree Basal Area	-	-	-	-	Canopy Cover (%)
Canopy Cover	**-**	**>70%**	**YES**	-	-
Snag Density	**Positive correlation**	-	**NO**	**Simon et al. 2002, White et al. 2016**	-

Data Collection

Bioacoustics data were collected by deploying remote recording devices (Song Meters; Wildlife Acoustics, Inc., Model SM2+). I designated song meter locations by randomly selecting locations within the study area under the criteria of reasonable accessibility (e.g., not in water bodies or on cliff faces) and a minimum distance between song meters of 300m. Three song meters were located within the boundary of the Scott Mountain Fire of 2010 while six were located in unburned forest (Figure 1). Song meters recorded in ten-minute intervals from late June to mid-July every morning at 4:45am, 5:30am, 6:15am, 7:15am, and 8:15am between the years 2014 and 2016. This resulted in ~4200 recordings over the span of three years, accounting for song meter malfunctions.

I selected a subsample of the 4,200 recordings between the years 2014 to 2016 for analysis using audio and spectrogram software Audacity (Audacity Version. 2.3.1). I randomly selected one recording per song meter per week during the peak of breeding season (6/30 - 7/26) for each year. The randomly selected day sampled for each week was used for all song meters of the same year; however I randomized the recording time sampled for each song meter. I selected six weeks of samples each year, resulting in 18, ten-minute recordings analyzed for each song meter. However due to song meter equipment failure and weather conditions prohibiting the deployment of recording devices, bioacoustics data collection varied between plots, resulting in a total of 1,160 minutes of evaluated recordings over the course of the study. In recordings where species detections were inaudible due to environmental

factors such as excessive rain or wind, the recording 24 hours before or after was sampled, as determined by a coin flip. Recorded songs, calls, and other sounds associated with avian activity from this subsample were identified to species by either Dr. Joan Hagar or myself. Detections of hermit and varied thrushes were cataloged as either being present in recordings (1) or not being present in recordings (0).

I used the documented vegetation and fuels characteristics collected in a companion study (Acker et al. unpublished). Vegetation data were collected at each of the nine song meter locations as well as at 10 additional locations (for a total of three locations in the Scott Mountain burn and 16 locations in the unburned area; Figure 1) during the years 2016 and 2018. Both tree data and shrub/forb data were collected using the tree sampling method and the line intercept sampling method outlined in the FIREMON (Fire Effects Monitoring and Inventory System) manual (Lutes et al. 2006). I categorized "fruit-bearing shrub" as any shrub species that produces fleshy fruit that would be consumed by local wildlife in the region (Appendix Table 3).

Data Analysis

For each year recordings were sampled, the detection frequency of each species in each plot was calculated by dividing the number of recordings the focal species was present by the number of recordings sampled (n = 6). The detection frequency of each plot of the same habitat type (burned or unburned) was then averaged across between all years that data was collected, generating a naïve occupancy estimate (Ψ) for each species in burned and unburned conditions (The University of Vermont 2021). I used naïve occupancy estimates as my response variable to quantify the probability that a species was present in each habitat type. I used Welch's two sample T-tests to compare naïve occupancy estimates between burned plots (n = 3) and unburned plots (n = 6) to test for a difference in occurrence of each species between habitat conditions. I used an alpha value of 0.05 as my identifier of a statistically significant difference in naïve occupancy between burned and unburned conditions.

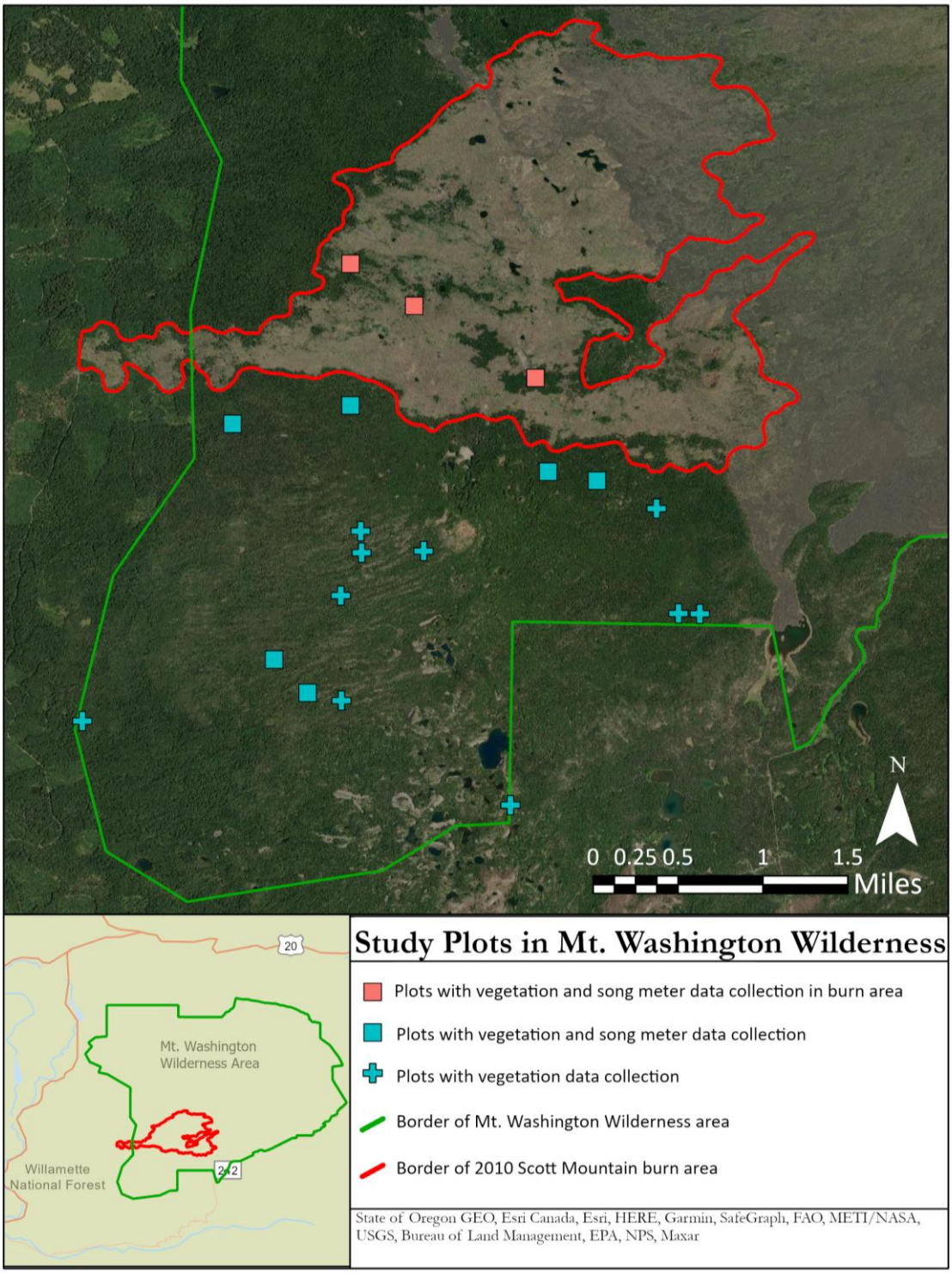
In addition to using T-tests, I also determined, a priori, a difference in estimated naïve occupancy between burned and unburned conditions that would indicate a potentially important ecological response in focal species occurrence. This additional analysis criterion was established in an effort to mitigate the small statistical power my T-test would have due to the small sample size used in this study. To establish an ecologically significant value, I calculated the difference between the maximum and minimum detection frequency of each species across all unburned plots. If the median

estimated naïve occupancy of the species in burned plots is less than the difference between the naïve occupancy estimate in unburned plots and the ecologically significant value, I concluded that there is not an ecologically significant difference in species occurrence between habitat types. If the median estimated naïve occupancy of the species in burned plots is greater than the difference between the naïve occupancy estimate in unburned plots and the ecologically significant value, I concluded that there is an ecologically significant difference in species occurrence between habitat types.

I used T-tests to compare each vegetation variable between burned plots (n = 3) and unburned plots, including plots where vegetation data, but not bioacoustics data was collected (n = 16). I compared fruit-bearing shrub cover (%), canopy cover (%), snag density (snags/plot), shrub cover (%), and live tree basal area (ft^2/acre). Because I was conducting multiple univariate comparisons between burned and unburned for each of the five vegetation variables, I applied a Bonferroni correction to the standard alpha value of 0.05 by dividing by the number of comparisons conducted (5 univariate T-tests), resulting in a new alpha value of 0.01.

Figure 1.) Map of Study Area

Map of song meter placement and vegetation data collection locations in the Mt. Washington Wilderness Area, Cascade Mountain range, Oregon.



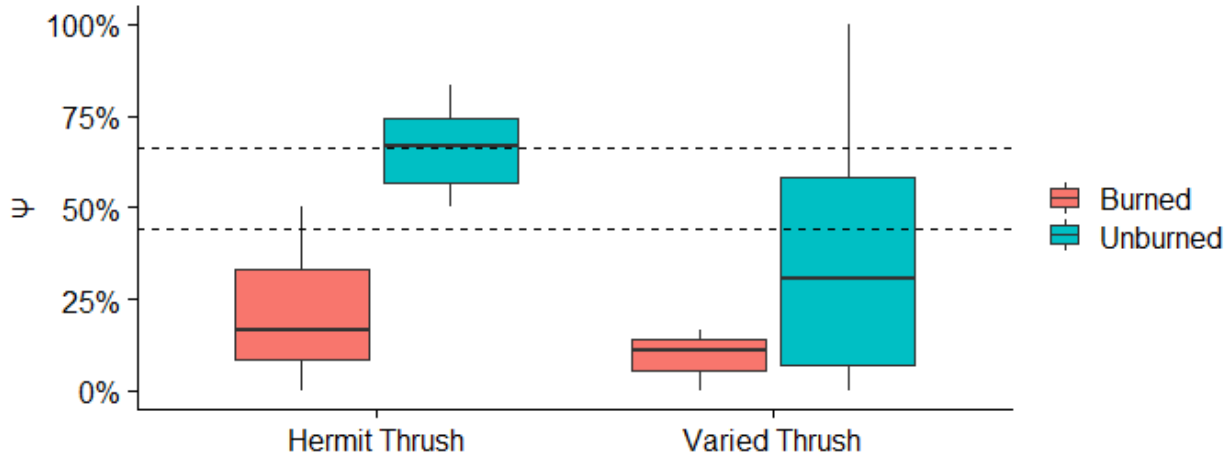
Chapter Three: Results

Hermit thrushes were detected in all unburned plots and two of the three burned plots (Appendix Table 1). Additionally, naïve occupancy of hermit thrushes was greater in unburned plots than in burned plots (Figure 2). Varied thrushes were detected in two of the three burned plots and four of the six unburned plots (Appendix Table 2). In general, hermit thrushes were detected at a higher rate and more consistently in both plot types than varied thrushes.

Ranges of naïve occupancy estimates for hermit thrush did not overlap between burned and unburned plots. Ranges of naïve occupancy estimates for varied thrush did overlap between plot types. For the hermit thrush, I used a 33% difference in estimated naïve occupancy between burned and unburned plots as my threshold for an ecologically significant difference in species occurrence because this was the range of estimated naïve occupancy in unburned plots for the species. For the varied thrush, I used a 100% difference in estimated naïve occupancy between burned and unburned plots as my threshold for an ecologically significant difference in species occurrence due to the 100% detection rate of the varied thrush in one unburned plot, and the 0% detection rate of the species in two unburned plots.

Figure 2.) Naive Occupancy (Ψ) Estimates for Thrush Species

Comparison of medians and ranges of estimated naïve occupancy (Ψ) for the hermit and varied thrush in burned and unburned plots in the Mt. Washington Wilderness area where bioacoustics data was collected. The ecologically significant difference in median estimated naïve occupancy between burned and unburned habitat of >33% for the hermit thrush is displayed on the plot as the horizontal dashed lines on the y axis. The ecologically significant value for the varied thrush is not displayed, but represents a 100% difference in median estimated naïve occupancy between burned and unburned habitats.



Hermit thrush estimated naïve occupancy satisfied the ecologically significant value of a >33% difference in detection frequency between burned and unburned plots. The ecologically significant value between habitat types for the varied thrush was 100%, and was not satisfied. Meeting the criteria for an ecologically significant difference in varied thrush occurrence between habitat types would have required a 100% detection frequency of the species in one habitat type and a 0% detection frequency of the species in the other habitat type. The difference in estimated naïve occupancy between burned and unburned plots for both focal species was not statistically significant (Table 2).

Table 2.) Avian T-test Results

Comparison of estimated naïve occupancy (Ψ) for two focal species between burned (n=3) and unburned (n=6) plots in the Mt. Washington Wilderness area where bioacoustics data were collected. Species T-test results are from Welch's two sample T-tests. Confidence was calculated using 95% confidence intervals. The standard alpha value of 0.05 was used and no adjustments for multiple comparisons were made.

	Burned Plots	Unburned Plots	T-Test Results					
	Mean ± (SE)	Mean ± (SE)	Burned - Unburned	Statistic	p value	Parameter	Conf.low	Conf.high
Species								
Hermit Thrush	22.22 ± (14.7)	66.2 ± (5.19)	-43.98	-2.82	0.082	2.51	-99.47	11.51
Varied Thrush	9.26 ± (4.9)	37.96 ± (16)	-28.70	-1.72	0.138	5.85	-69.90	12.49

One of the five vegetation characteristics that were compared in this study differed significantly between burned and unburned plots based on the Welch's two sample T-tests (Table 3). Mean live tree basal area was 181 ft^2/ac greater (95% CI 141 - 221ft^2/ac greater) in unburned than in burned plots.

Median live tree basal area was greater in unburned plots than in burned plots by 166% (Figure 3). A documented prescription of live tree basal area amount suitable for hermit and varied thrush habitat in the high Cascades was not identified during the literature review portion of this study. However, all plots with greater than 25% detection frequency of varied thrush had a live tree basal area greater than 180 ft^2/ac (Figure 8 – C). Plots with the highest detection frequencies of hermit thrush also tended to have the highest basal area and canopy cover, but these two variables were highly correlated with whether the plot was burned or unburned (Figure 9 – C & D).

Figure 3.) Live Tree Basal Area in Burned and Unburned Plots

Comparison of average live tree basal area (ft^2/acre) between all burned and unburned plots in the Mt. Washington Wilderness, Oregon Cascades.

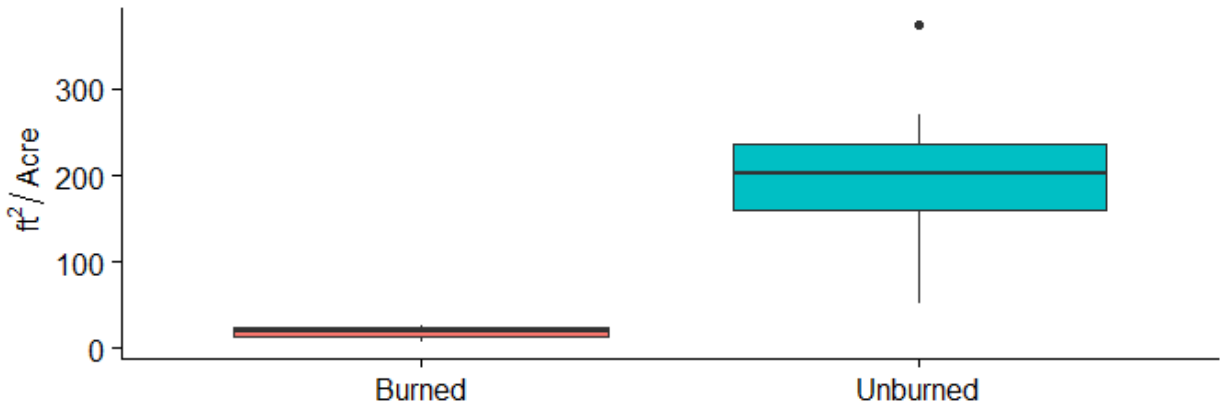


Table 3.) Vegetation T-test Results

Comparisons of vegetation characteristics between burned and unburned plots in the Mt. Washington Wilderness area. Mean and standard error (SE) summaries for burned (n=3) and unburned (n=16) plots are presented first, followed by Welch's two sample T-test results. Confidence was calculated using 95% confidence intervals. The standard alpha value of 0.05 was divided by the number of T-test comparisons being conducted (5), resulting in a new alpha value of 0.01. Vegetation characteristics important for either focal species as identified by Partners in Flight have a (*). Vegetation data characteristics that have been identified as important for either species by other sources in literature review have (**).

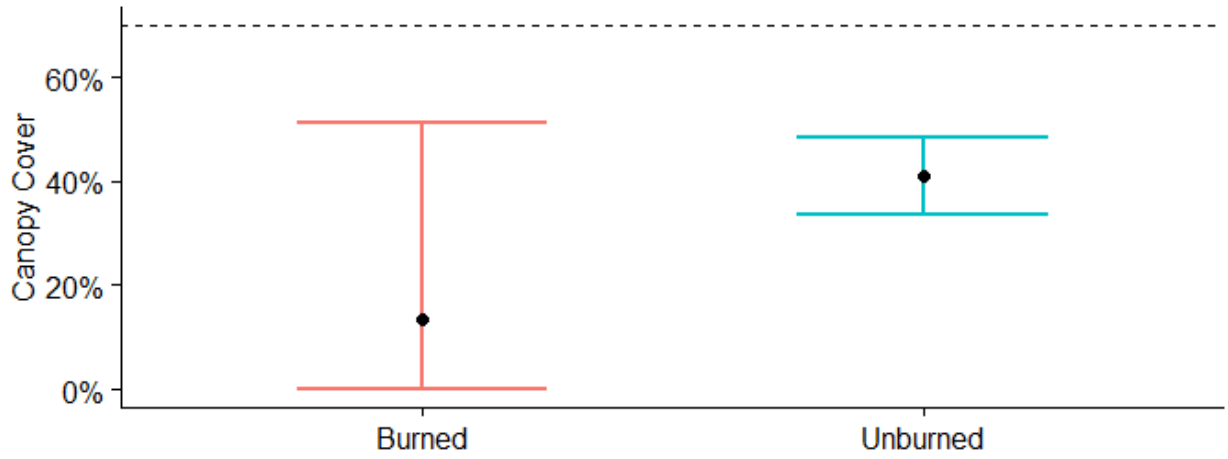
	Burned Plots	Unburned Plots	T-Test Results					
	Mean ± (SE)	Mean ± (SE)	Burned - Unburned	Statistic	p value	Parameter	Conf.low	Conf.high
Cover Percentages								
Fruit-Bearing Shrub Cover (%) *	10.98 ± (1.28)	12.78 ± (2.1)	-1.8	-0.73	0.68	29.74	-6.82	3.22
Total Shrub Cover (%) *	11.35 ± (1.13)	12.11 ± (1.88)	-0.76	-0.35	0.731	34.83	-5.21	3.69
Forest Stand Characteristics								
Canopy Cover (%) *	13.33 ± (8.82)	40.94 ± (3.51)	-27.60	-2.91	0.071	2.68	-59.99	4.79
Live Tree Basal Area (ft2/acre)	18.11 ± (5.88)	199.32 ± (17.97)	-181.20	9.58	< 0.0001	16.93	141.29	221.11
Snag Density (snag/acre by plot)								
Snag Density **	224.17 ± (43.16)	86.88 ± (12.96)	137.29	3.05	0.075	2.37	-30.08	304.67

I did not find a significant difference between burned and unburned plots for four of the five vegetation variables (canopy cover, snag density, shrub cover, and fruit-bearing shrub cover: Table 3). Previously documented requirements for suitable habitat for either focal species outlined in (Altman and Alexander 2012) were compared between plots types in order to determine whether or not they were satisfied.

The recommended canopy cover percentage of >70% for varied thrush (Altman and Alexander 2012) was not met in burned or unburned plots (Figure 4). A recommended canopy cover percentage for suitable hermit thrush habitat in the high Cascades was not identified during the literature review portion of this study. Although I did not find statistical evidence for a difference, mean canopy cover in unburned plots was 28% greater than the mean in burned plots (Table 3). Unburned plots had consistently greater canopy cover than burned plots, with low variability (Figure 4). The detection frequency of hermit thrush was highest on unburned plots with intermediate canopy cover (30-45%) (Figure 9 – D) but the low sample size makes the relationship difficult to evaluate. Within burned plots, there was a weak positive trend of the plots with the highest varied thrush detection frequencies also having the highest canopy cover (Figure 8 – D). The association of varied thrush detection frequency with canopy cover was difficult to assess since there weren't any plots within the recommended threshold for canopy cover.

Figure 4.) Canopy Cover in Burned and Unburned Plots

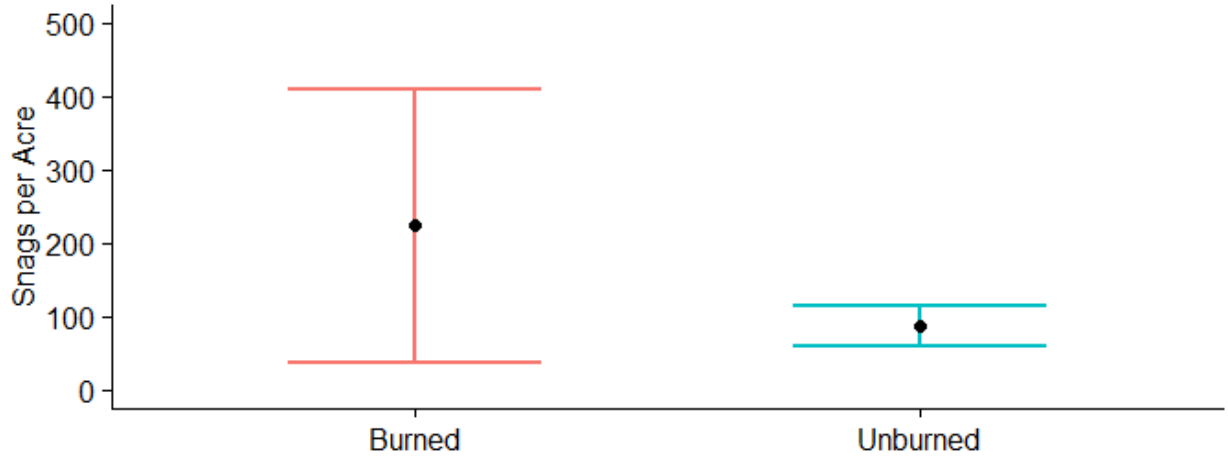
Comparison of mean and confidence intervals for canopy cover between all burned and unburned plots in the Mt. Washington Wilderness, Oregon Cascades. The recommended canopy cover documented in (Altman and Alexander 2012) of 70% or greater for the varied thrush is displayed on the plot as the horizontal dashed line**.**



A recommended density of snags for suitable hermit and varied thrush habitat in the high Cascades was not identified during the literature review portion of this study. However, snag density has been positively associated with the hermit thrush in previous studies outside of the high Cascades (Schwab et al. 2001, Simon et al. 2002, White et al. 2016). Although I did not find statistical evidence for a difference, mean snag density in burned plots was 88% greater than the mean in unburned plots (Table 3). Burned plots had a higher density of snags with high variability while unburned plots had a lower density of snags with more consistency (Figure 5). Within burned and unburned plots separately, there was a weak positive trend of the plots with the highest hermit thrush detection frequencies also having the highest snag densities (Figure 9 - E) supporting the previously documented association between hermit thrush occurrence and snag density.

Figure 5.) Snag Density in Burned and Unburned Plots

Comparison of mean and confidence intervals for density of snags between all burned and unburned plots in the Mt. Washington Wilderness, Oregon Cascades.

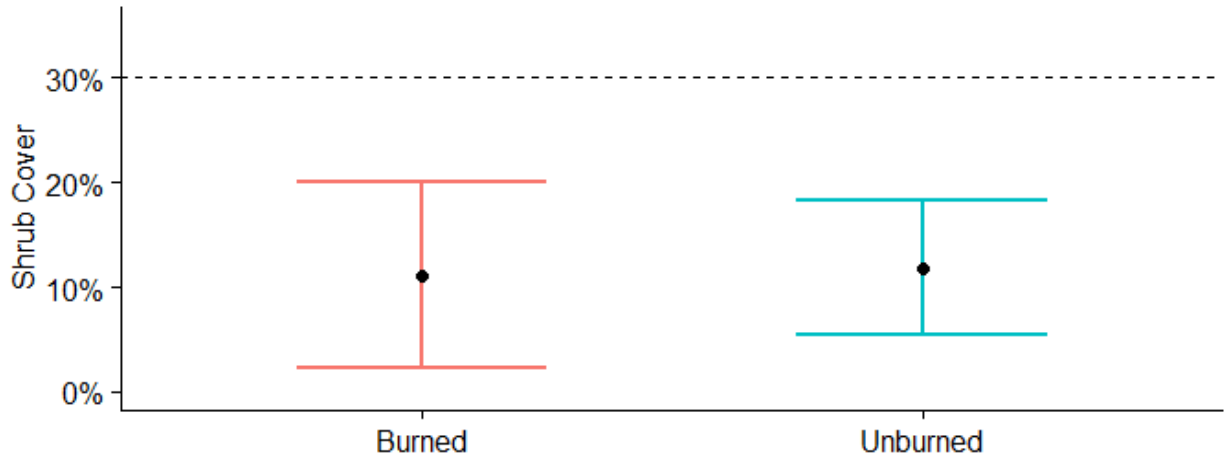


I did not find a significant difference in shrub cover between burned and unburned plots. The mean shrub cover in both unburned and burned plots was well below the recommended shrub cover percentage of 30-70% for the hermit thrush habitat and the recommended sub-canopy cover amount of >30% for suitable varied thrush habitat (Altman and Alexander 2012) was not met in burned or unburned plots (Figure 6). Variability in shrub cover between burned and unburned plots were similar to one another. All plots with a detection frequency of varied thrush greater than 25% had a shrub cover percent of <5% (Figure 8 – B). My results do not coincide with the previously documented recommendations for shrub cover of both species in each plot type, revealing a lack of difference in

shrub cover amount in burned and unburned plots, as well as the occurrence of both thrush species in plots below the recommended cover percent for each species.

Figure 6.) Shrub Cover in Burned and Unburned Plots

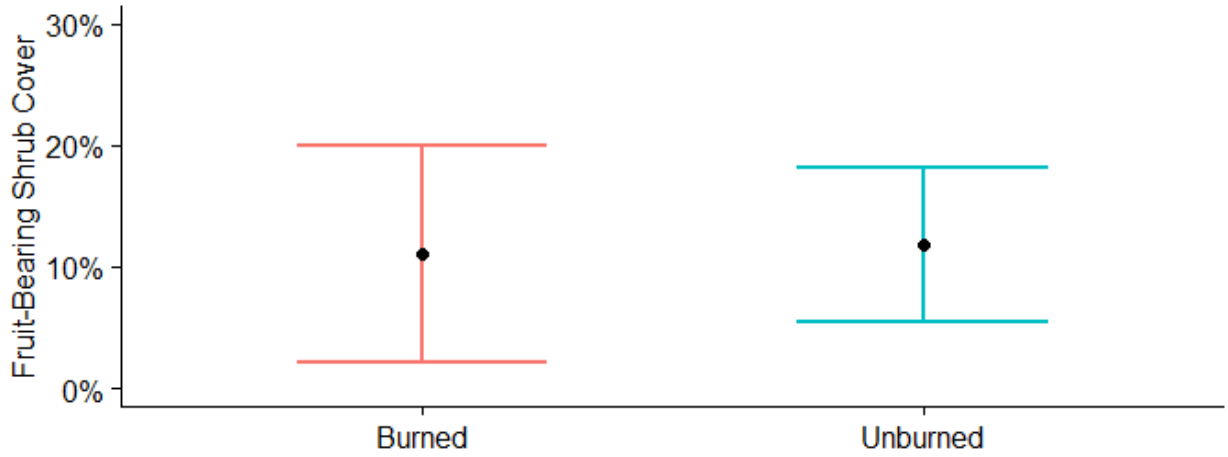
Comparison of mean and confidence intervals for shrub cover between all burned and unburned plots in the Mt. Washington Wilderness, Oregon Cascades. The recommended minimum range of total shrub cover as documented in (Altman 2012) of 30% - 70% for the hermit thrush and the recommended minimum requirement of sub-canopy cover as documented in (Altman 2012) of >30% for the varied thrush is displayed on the plot as the horizontal dashed line.



A previously documented amount of fruit-bearing shrub cover suitable for hermit and varied thrush habitat in the high Cascades was not identified during the literature review portion of this study. However, fruit-bearing shrub cover has been documented as an important habitat feature for varied thrush in Oregon Partners in Flight (Altman and Alexander 2012). Although I did not find statistical evidence for a difference, variability in shrub cover was similar between burned and unburned plots (Figure 7).

Figure 7.) Fruit-Bearing Shrub Cover in Burned and Unburned Plots

Comparison of mean and confidence intervals for fruit-bearing shrub cover between all burned and unburned plots in the Mt. Washington Wilderness, Oregon Cascades.



Similar to shrub cover, plots with a detection frequency of varied thrush greater than 25% had a fruit-bearing shrub cover percent of <5% (Figure 8 – A). A correlation between hermit thrush detection frequency and fruit-bearing shrub cover was not observed (Figure 9 – A). Fruit-bearing shrub species that are associated with hermit and varied thrush diets were found in both plot types where bioacoustics data was collected. *Vaccinium* species were found in both burned and unburned plots in similar cover percentages; however the range of species in the genus varied between plot types (Table 4). *Ribes viscosissimum* was found in one burned plot, but not in any unburned plots. Finally, *Arctostaphylos nevadensis* was found in one burned plot, but not in any unburned plots. In absence of a recommended fruit-bearing shrub cover amount for either species, and without evidence of an observed correlation between varied thrush detection frequency and fruit-bearing shrub cover, I did not find evidence of an association of thrushes with fruit-bearing shrub cover.

When comparing correlations between species detection frequencies and vegetation characteristics, several relationships between vegetation variables were observed. In burned plots, an inverse relationship between shrub cover and live tree basal area was observed (Figure 8 & 9 – B & C). In unburned plots, a general positive association between shrub cover and snag density was observed (Figure 8 & 9 – B & E). Finally, an inverse relationship between live tree basal area and snag density was observed in both plot types (Figure 8 & 9 – C & E).

Table 4.) Comparison of Shrub Species Cover Between Plot Types

Comparison of fruit-bearing and non-fruit-bearing shrub cover by species in burned and unburned plots where bioacoustics data was collected. Purple cells resemble shrub species that are fruit-bearing while orange cells resemble shrub species that are not fruit-bearing. *Vaccinium and Arctostaphylos* have been documented as part of the diet of the varied thrush in the high Cascades. *Ribes* has been documented as part of the diet of the hermit thrush in the high Cascades.

	Burned Plots		Unburned Plots	
	Mean (%)	Range (%)	Mean (%)	Range (%)
Shrub Species				
Arctostaphylos nevadensis	1.36	1.36	-	-
Gaultheria ovatifolia	-	-	0.96	0.96
Pachistima myrsinites	0.35	0.35	-	-
Ribes viscosissimum	1.06	1.06	-	-
Vaccinium membranaceum	7.22	4.60 - 11.50	5.24	0.25 - 19.19
Vaccinium scoparium	8.84	8.84	10.01	0.20 - 37.07

Figure 8.) Varied Thrush and Vegetation Characteristics Comparisons

Relationships of varied thrush detection frequencies with vegetation characteristics in burned and unburned plots.

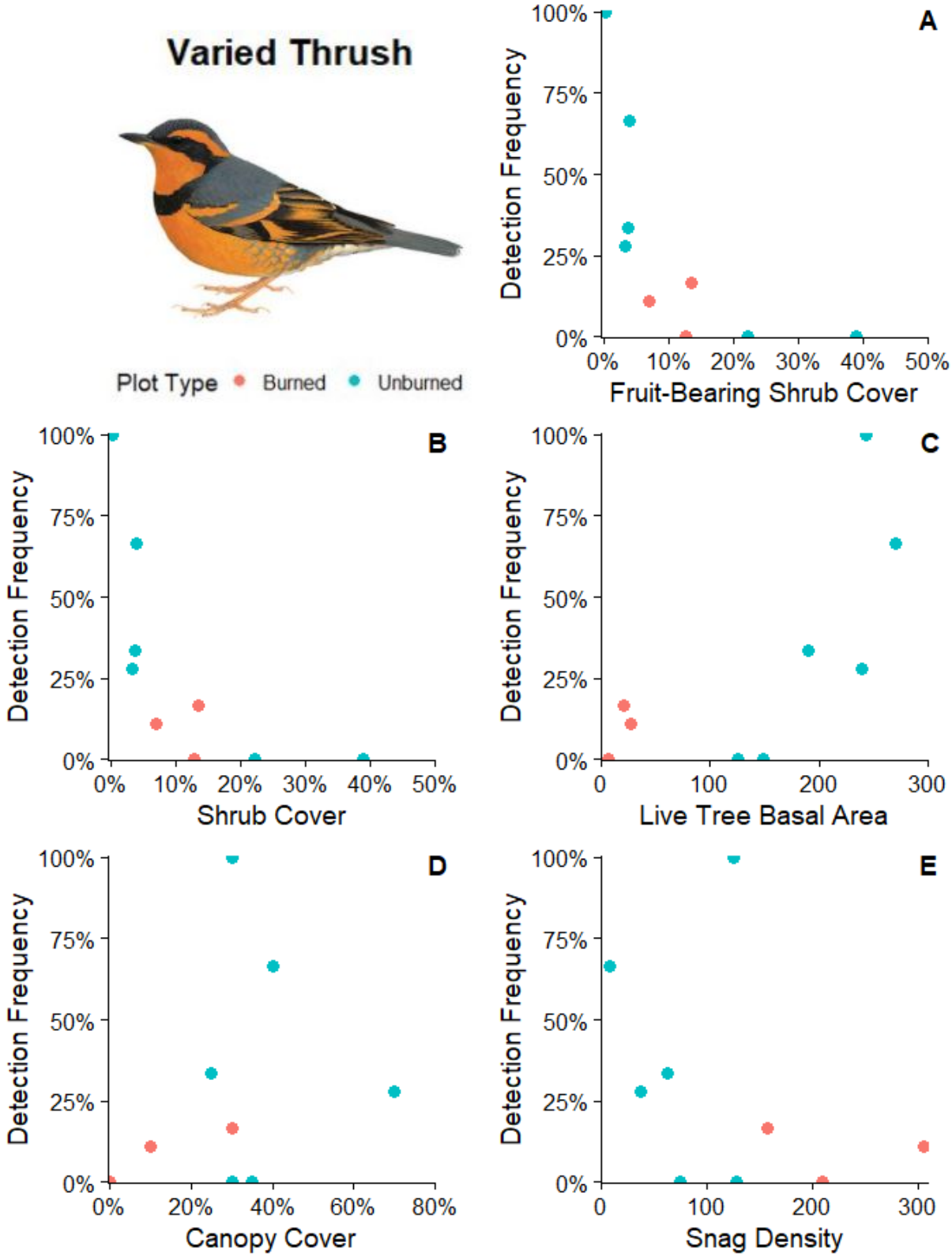
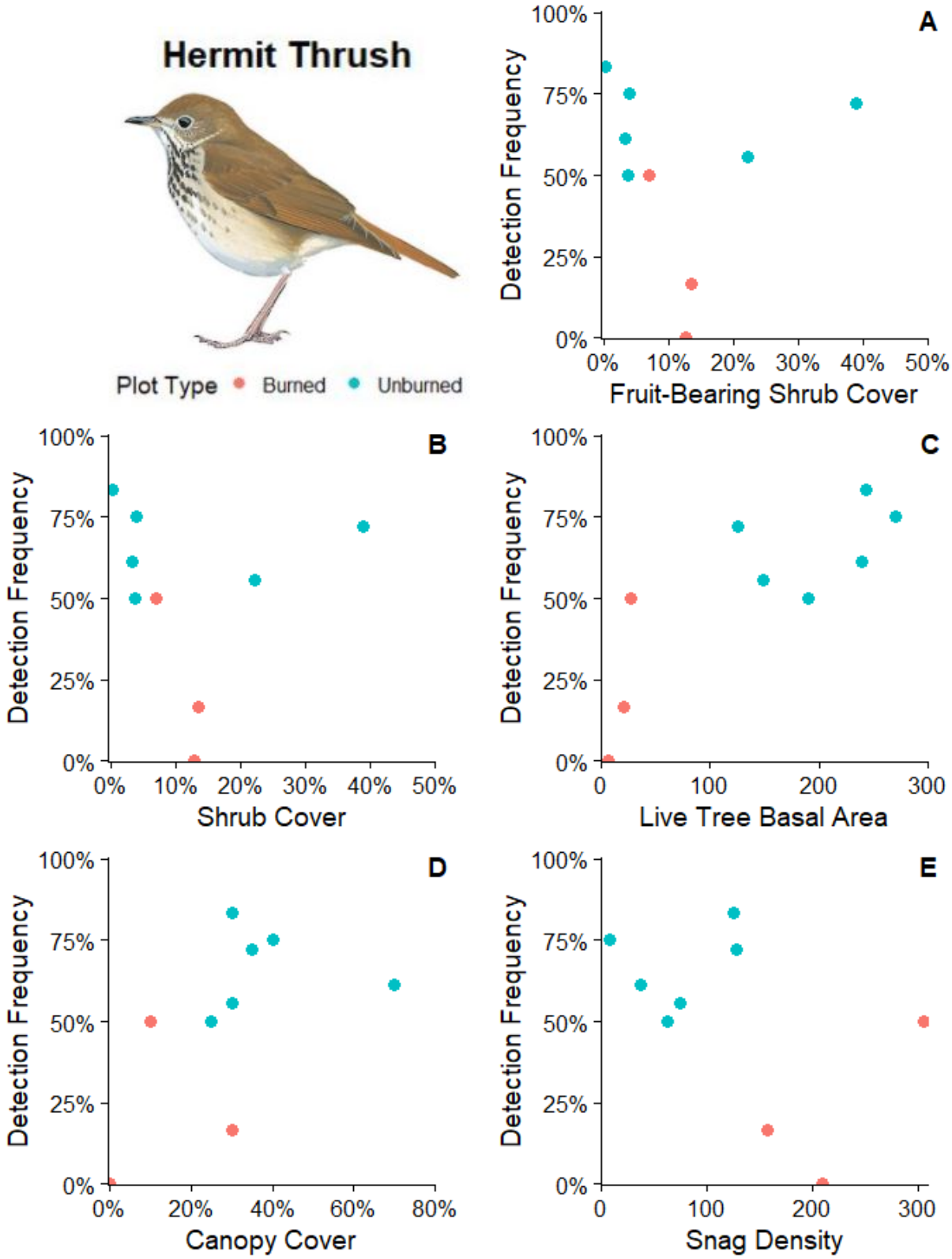


Figure 9.) Hermit Thrush and Vegetation Characteristics Comparisons

Relationships of hermit thrush detection frequencies with vegetation characteristics in burned and unburned plots.



Chapter Four: Discussion

While the two thrush species occupy similar forest types in the high Cascades, hermit and varied thrushes were associated with different habitat characteristics sampled in this study, indicating a potentially different response to fire disturbance. Differences in vegetation characteristics may have influenced the occurrence of hermit thrush in burned and unburned plots. However, occurrence of varied thrush was not strongly associated with burn condition, but was strongly associated with live tree basal area. Both species were detected during the breeding season in the high Cascades in unburned and burned plots despite the previously documented association of varied thrush with late successional forests. The low detection frequency of the varied thrush across all plots may have been affected by the sub-optimal habitat characteristics present in each habitat, compared to the habitat characteristics outlined by Oregon Partners in Flight or due to population declines of the species in the high Cascades. The uneven sample size of burned and unburned plots as well as the small sample size in general clearly influenced the statistical power of this study. However, my results contribute to a better understanding of hermit and varied thrush distribution patterns after fire in western forests, and can help guide future research on the effects of fire disturbance on these species and their habitat.

Hermit Thrush

Snag density has been positively correlated with hermit thrush in previous studies outside of the high Cascades (Schwab et al. 2001, White et al. 2016). Although hermit thrushes have been hypothesized to use snags as cover, singing posts, or as sources of food (invertebrates) (Simon et al. 2002), this biological association is not clearly supported by evidence in previous studies. Additionally, the removal of snags through salvage logging following a fire has been associated with a sudden reduction in the number of hermit thrushes in the past (LeCoure 1999) although the exact cause of this is unknown. It is possible that snags in burned plots fulfill the ecological role that live trees typically fulfill as singing posts for hermit thrushes during the breeding season, and/or that snags are a source of invertebrate prey for hermit thrush.

While I did not find evidence of a positive correlation between hermit thrush occurrence and snag density, the relationship between hermit thrush occurrence and snag densities was consistent between burned and unburned plots. Previous studies have found insect composition to be similar across burned plots but to differ significantly among unburned plots (Wikars 2002). A greater variation of insects in unburned habitat may provide greater opportunities for foraging hermit thrushes.

Additionally, snag formation may have a positive correlation with occurrence of cavity-nesting species, with unburned snags providing shelter for cavity-nesters for a greater period of time than burned snags (Edworthy et al. 2012). It is possible that the method of snag formation could affect a non-cavity nesting species such as the hermit thrush, though specific reasons as to how are largely unknown. Future studies on hermit thrush interactions with snags in burned and unburned habitat in the high Cascades could increase understanding of the effects that fire disturbance may have on hermit thrushes and their habitat.

In two of three burned plots, the shrub cover was less than the ideal amount for hermit thrush suggested in Altman and Alexander (2012), yet hermit thrushes were detected. In the burned plot where hermit thrush were not detected, vegetation cover was low in both shrub and canopy form. Previous literature has identified the hermit thrush's generalist nature for forest habitat, with the species documented in burned (Schulte and Niemi 1998) and unburned forest habitat (Jones and Donovan 1996, Altman and Alexander 2012). However, a high severity burn that leaves little to no cover, as demonstrated in the burned plot without detection of the hermit thrush, may not provide adequate habitat for the species in the high Cascades.

Live tree basal area has not been used to quantify hermit thrush habitat, but provides an additional metric that can be used to assess habitat availability for the hermit thrush. Hermit thrushes were detected in greater frequency in burned plots that retained more live trees following the Scott Mountain Fire, suggesting that the retention of live trees following a fire disturbance may be beneficial for the species. Severe fires in the high Cascades have the potential to burn entire stands of forest, reducing the number of live trees in the area while affecting landscape structure and composition (Spies 2018) and may not provide suitable habitat for the hermit thrush. Conversely, low severity fire has the potential to create structure and habitat conditions that are favorable for many avian species (Stephens et al. 2012), including the retention of some live trees and canopy cover (Tepley et al. 2013), increasing fruit-bearing shrub abundance (Minore 1972), creating beneficial foraging conditions for avian species (Beck and George 2000), and increasing snag densities (Kennedy and Fontaine 2009). Fire severity may have an impact on hermit thrush habitat, and should be researched further in future studies with the inclusion of live tree basal area as a metric for assessing habitat.

I have concluded that the occurrence of hermit thrushes in the Mt. Washington Wilderness area is greater in unburned habitats than in burned habitats. However, reasons for the possible variation in occurrence of the species in burned plots should be studied further, with the integration of fire severity

serving as the focal point of future research. Variation in fire severity may affect hermit thrush occurrence by providing varying conditions in both quality and type of habitat characteristics that are present following a fire disturbance in the high Cascades. Additionally, the successional development of habitat following a wildfire disturbance could have an effect on species occurrence, with the establishment of cover, canopy, and live trees providing suitable habitat for hermit thrush as the time since fire progresses.

Recurring mixed severity fire serves an important function in sustaining ecological processes such as the creation of a mosaic of seral stages and habitats structures (Agee 1996). Recurring mixed severity fire can be reintroduced in wilderness areas through the use of prescribed fire, although support from the public and federal funding for prescribed burning in the Pacific Northwest needs to be increased to do so effectively (Wimberly and Liu 2014). In addition to reducing the size and number of wildfires, and their severity during the fire season, the implementation of prescribed fires provides the opportunity for ecological experiments to be conducted, furthering our understanding of the role and impacts that fire disturbance has on individual species, such as the hermit thrush as well as entire communities of fauna in the western United States.

Varied Thrush

In unburned areas, plots with the lowest amount of shrub and fruit-bearing shrub cover had the greatest detection frequencies of varied thrush. The observed negative association between varied thrush occurrence and shrub cover was unexpected, given the importance that shrub cover holds for the species (Altman and Alexander 2012). Several explanations for this observed relationship can be proposed based on documented diet and foraging characteristics of varied thrushes during the breeding season or by fruit development and availability of shrub species in the high Cascades.

Varied thrushes are associated with late successional or old growth forest (Altman and Alexander 2012) that provide high amounts of shrub cover for nesting, foraging, and shelter (George 2000). However, previous studies have questioned whether or not varied thrushes benefit from areas with low shrub cover, proposing that invertebrates are more easily detected in areas with less shrub cover, making prey more visible to foraging birds (Beck and George 2000). I did not address abundance of invertebrate prey in my study, or its potential influence on varied thrush occurrence. It is possible that there was a greater opportunity for varied thrush to detect invertebrates in unburned plots with low shrub cover. In effort to understand the effects that food abundance has on varied thrush

occurrence, future studies of the species in the high Cascades could prioritize the sampling of invertebrate prey in addition to other vegetation variables related to varied thrush diet.

The diet of the varied thrush changes from invertebrates to fruit as the breeding season progresses (George 2000) and fruits ripen. Therefore, fruit may not be an integral part of the varied thrush diet during the early part of the breeding season, and may become more important after the breeding season. Fruit production by *Vaccinium* species in climates similar to that in this study can be affected by meteorological events such as snow pack amount and duration (Minore 1972). In addition, pollination rates may be negatively affected by cold or wet weather (Friesen 2016) and may influence the availability of berries for varied thrushes. It is possible that the observed negative correlation between fruit-bearing shrub cover and varied thrush occurrence could have been affected by fruit availability or that the availability of fruit from shrubs is restricted by other habitat characteristics such as shade from canopy cover. In future studies, a sampling of fruit production could more accurately reflect food potential for frugivores. However sampling of genera such as *Vaccinium* would take several years in order to gain an accurate average measurement of berry production and time of ripening within each study area (Minore 1972).

Varied thrush occurrence had a positive relationship with canopy cover and live tree basal area in burned plots following the Scott Mountain Fire. A positive correlation between the species and canopy cover was expected based on associations of the species with forests in a late successional period (Altman and Alexander 2012). Similar to the hermit thrush, a relationship between the varied thrush and live tree basal had not been studied, and provides an alternative variable for assessing varied thrush habitat. My results show that varied thrush occurred in plots that retained live trees following the fire disturbance, indicating that fire severity may have an impact on varied thrush habitat within burn areas. The lack of evidence for a difference in varied thrush occurrence between burned and unburned plots suggests that both plot types provided habitat for the species. An increased sample size of burned and unburned plots in future studies and inclusion of fire severity measurements as predictor variables could further inform factors influencing varied thrush occurrence in pre- and post-fire landscapes.

Chapter Five: Conclusion

I found that hermit thrush occurrence differed between burned and unburned plots in the Mt. Washington Wilderness, suggesting that forest fires may influence availability of their habitat. The

difference in hermit thrush occurrence between burned and unburned plots was not associated with any of the key habitat features to which hermit thrush were expected to respond. Contrary to expectation, I did not find evidence of a positive correlation between snag density or shrub cover with hermit thrush occurrence.

I did not find evidence for a difference in varied thrush occurrence between burned and unburned plots. However, among unburned plots, varied thrush occurrence was greatest where shrub cover was least. Possible explanations include variation in fruiting and variation in availability of insect prey. Although varied thrushes are primarily associated with late successional forests (George 2000, Marshall et al. 2003, Altman and Alexander 2012), they still occurred in burned areas where trees remained.

Finally, I found a positive correlation between live tree basal area and the occurrence of both thrush species, providing an additional metric that can be used to assess habitat availability that has not been identified previously. The retention of live trees following a fire disturbance can provide adequate habitat for both species in the high Cascades.

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Appendices

Appendix Table 1.) Hermit Thrush Detection Frequency

Detection frequency of hermit thrush at each plot between 2014 and 2016. Blue plots are unburned and red plots are located within the 2010 Scott Mtn. fire perimeter. Green values represent the detection of the species during the sampled recording, and purple values represent a mechanical or recording error of the sampled recording. Det. % represents the detection frequency of the species across all sampled recordings.

Detection Data																			
Hermit thrush (*Catharus guttatus*)																			
Meter	2014 Survey						2015 Survey						2016 Survey						Det. %
-	S1	S2	S3	S4	S5	S6	S1	S2	S3	S4	S5	S6	S1	S2	S3	S4	S5	S6	-
Plot 5	0	-	-	-	0	-	-	-	-	-	-	-	0	0	0	0	0	0	0%
Plot 6	0	1	0	0	0	0	1	1	1	1	1	1	0	0	1	1	0	0	50%
Plot 9	-	-	-	-	-	-	-	-	-	-	-	-	0	1	0	0	0	0	17%
Plot 3	1	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	72%
Plot 4	1	0	0	1	1	1	0	1	1	1	0	0	1	1	1	1	0	0	61%
Plot 7	1	1	1	1	1	0	-	-	-	-	-	-	-	-	-	-	-	-	83%
Plot 10	0	1	0	1	1	1	1	1	0	1	0	0	1	1	0	1	0	0	56%
Plot 11	1	1	0	1	1	1	-	-	-	-	-	-	1	1	1	0	1	0	75%
Plot 12	1	1	1	0	0	0	0	1	1	0	0	1	-	-	-	-	-	-	50%

Appendix Table 2.) Varied Thrush Detection Frequency

Detection frequency of varied thrush at each plot between 2014 and 2016. Blue plots are unburned and red plots are located within the 2010 Scott Mtn. fire perimeter. Green values represent the detection of the species during the sampled recording, and purple values represent a mechanical or recording error of the sampled recording. Det. % represents the detection frequency of the species across all sampled recordings.

Detection Data																			
Varied thrush (*Ixoreus naevius*)																			
Meter	2014 Survey						2015 Survey						2016 Survey						Det. %
-	S1	S2	S3	S4	S5	S6	S1	S2	S3	S4	S5	S6	S1	S2	S3	S4	S5	S6	-
Plot 5	0	-	-	-	0	-	-	-	-	-	-	-	0	0	0	0	0	0	0%
Plot 6	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	11%
Plot 9	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	1	0	0	17%
Plot 3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0%
Plot 4	0	1	1	0	0	0	1	0	0	0	0	0	1	1	0	0	0	0	28%
Plot 7	1	1	1	1	1	1	-	-	-	-	-	-	-	-	-	-	-	-	100%
Plot 10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0%
Plot 11	1	1	1	1	1	1	-	-	-	-	-	-	1	0	0	1	0	0	67%
Plot 12	0	0	0	0	1	1	1	0	0	0	1	0	-	-	-	-	-	-	33%

Appendix Table 3.) Habitat Characteristics of Each Plot Where Bioacoustics Data Was Collected

Habitat characteristics of each plot where bioacoustics data was collected, followed by averages of values by plot type. Purple cells resemble shrub species that are fruit-bearing while orange cells resemble shrub species that are not fruit-bearing.

	Burned Plots			Unburned Plots						Average	
	Plot 5	Plot 6	Plot 9	Plot 3	Plot 4	Plot 7	Plot 10	Plot 11	Plot 12	Burned	Unburned
Canopy Cover (%)	**0**	**10**	**30**	**35**	**70**	**30**	**30**	**40**	**25**	**13.33**	**38.33**
Live Basal Area (ft2/acre)	**6.78**	**26.51**	**21.04**	**124.95**	**239.36**	**243.23**	**149.06**	**270.09**	**190.29**	**18.11**	**202.83**
Total Snag Density per Acre	**210.0**	**305.0**	**157.5**	**127.5**	**37.5**	**125.0**	**75.0**	**7.5**	**62.5**	**224.17**	**72.50**
Snag Density of Trees 4"-10" DBH	70.00	270.00	60.00	90.00	-	80.00	30.00	-	10.00	133.33	52.50
Snag Density of Trees 10"-20" DBH	132.50	32.50	80.00	35.00	32.50	40.00	40.00	2.50	42.50	81.67	32.08
Snag Density of trees > 20" DBH	7.50	2.50	17.50	2.50	5.00	5.00	5.00	5.00	10.00	9.17	5.42
Shrub Cover (%)	**12.88**	**6.97**	**13.43**	**39.09**	**3.18**	**0.25**	**22.22**	**3.94**	**3.74**	**11.09**	**12.07**
Fruit-Bearing Shrub Cover (%)	**12.53**	**6.97**	**13.43**	**39.09**	**3.18**	**0.25**	**22.22**	**3.94**	**3.74**	**10.98**	**12.07**
Arctostaphylos nevadensis	-	1.36	-	-	-	-	-	-	-	1.36	0.00
Gaultheria ovatifolia	-	-	-	-	-	-	0.96	-	-	0.00	0.96
Pachistima myrsinites	0.35	-	-	-	-	-	-	-	-	0.35	0.00
Ribes viscosissimum	1.06	-	-	-	-	-	-	-	-	1.06	0.00
Vaccinium membranaceum	11.46	5.61	4.59596	2.02	2.47	0.25	19.19	3.74	3.74	7.22	5.24
Vaccinium scoparium	-	-	8.838384	37.07	0.71	-	2.07	0.20	-	8.84	10.01

Appendix Table 4.) Shrub Cover Percent of Every Plot

Shrub cover percent and averages of each species found in burned and unburned plots. Plots in the left two columns that are colored red represent burned plots while plots colored blue represent unburned plots where vegetation data was collected. Averages of each plot type and each individual species is reported in the last two rows. Purple cells denote shrub species that are fruit-bearing while orange cells indicate shrub species that are not fruit-bearing.

	Total Cover	*A. nevadensis*	*G. humifusa*	*G. ovatifolia*	*P. myrsinites*	*R. viscosissimum*	*Rosa gymnocarpa*	*V.deliciosum*	*V. membranaceum*	*V. ovalifolium*	*V. scoparium*
Plot 5	12.88	-	-	-	0.35	1.06	-	-	11.46	-	-
Plot 6	6.97	1.36	-	-	-	-	-	-	5.61	-	-
Plot 9	13.43	-	-	-	-	-	-	-	4.60	-	8.84
Plot 3	39.09	-	-	-	-	-	-	-	2.02	-	37.07
Plot 4	3.18	-	-	-	-	-	-	-	2.47	-	0.71
Plot 7	0.25	-	-	-	-	-	-	-	0.25	-	-
Plot 10	22.22	-	-	0.96	-	-	-	-	19.19	-	2.07
Plot 11	3.94	-	-	-	-	-	-	-	3.74	-	0.20
Plot 12	3.74	-	-	-	-	-	-	-	3.74	-	-
Plot 14	0.66	-	-	-	-	-	-	-	0.35	-	0.30
Plot 15	7.27	-	-	-	-	-	-	-	7.27	-	-
Plot 16	4.39	-	-	-	-	-	-	-	2.12	-	2.27
Plot 17	4.80	-	-	-	0.10	-	-	-	4.70	-	-
Plot 18	35.96	-	-	-	-	-	-	-	35.96	-	-
Plot 19	9.39	-	-	-	-	-	-	-	9.39	-	-
Plot 20	11.36	0.96	-	-	0.10	-	-	-	10.30	-	-
Plot 21	9.19	-	-	-	-	-	-	1.97	6.77	0.45	-
Plot 22	6.01	-	-	-	0.40	-	0.05	0.00	5.56	-	-
Plot 23	19.70	-	0.15	-	-	-	-	3.48	16.06	-	-
Average											
Burned	**11.09**	1.36	0.00	0.00	0.35	1.06	0.00	0.00	7.22	0.00	8.84
Unburned	**11.32**	0.96	0.15	0.96	0.20	0.00	0.05	1.82	8.12	0.45	7.10

Appendix Table 5.) Overstory Characteristics of Every Plot

Habitat characteristics of every plot where vegetation data was collected, followed by averages of values by plot type.

	Canopy Cover (%)	Live Basal Area (# of Trees/Plot)	Total snag Density per Plot	Snag Density of Trees 4"-10" DBH	Snag Density of Trees 10"-20" DBH	Snag Density of trees > 20" DBH
Plot 5	0	6.78	210.00	70.00	132.50	7.50
Plot 6	10	26.51	305.00	270.00	32.50	2.50
Plot 9	30	21.04	157.50	60.00	80.00	17.50
Plot 3	35	124.95	127.50	90.00	35.00	2.50
Plot 4	70	239.36	37.50	-	32.50	5.00
Plot 7	30	243.23	125.00	80.00	40.00	5.00
Plot 10	30	149.06	75.00	30.00	40.00	5.00
Plot 11	40	270.09	7.50	-	2.50	5.00
Plot 12	25	190.29	62.50	10.00	42.50	10.00
Plot 14	60	374.67	75.00	50.00	22.50	2.50
Plot 15	40	204.70	20.00	-	7.50	12.50
Plot 16	60	235.57	25.00	-	7.50	17.50
Plot 17	50	162.36	180.00	110.00	65.00	5.00
Plot 18	45	205.71	162.50	110.00	47.50	5.00
Plot 19	30	171.83	97.50	30.00	60.00	7.50
Plot 20	20	51.29	150.00	80.00	67.50	2.50
Plot 21	40	137.96	90.00	80.00	7.50	2.50
Plot 22	50	198.47	95.00	20.00	62.50	12.50
Plot 23	30	229.55	60.00	30.00	27.50	2.50
Average						
Burned	13.33	18.11	224.17	133.33	81.67	9.17
Unburned	40.94	199.32	86.88	60.00	35.47	6.41